

g LX Version 9.0 Applications Update

Dear Customer,

Congratulations! Your LX system is scheduled to receive a Version 9.0 FMI software installation. This FMI will include a change to the GRx user interface, pulse sequence revisions, changes to workflow in several applications, and multiple new system features.

This document has been provided to communicate important information regarding the changes and features included with the Version 9 FMI, and to share applications tips and suggestions you and your colleagues will find helpful during the initial use of this new software.

It is strongly recommended that you thoroughly review this introductory information, in conjunction with the Version 9.0 and ASP2 operator documentation included with the FMI kit prior to scheduling your software installation. Taking the time to review these materials in advance will ensure that you obtain optimal performance from your new software, and promote a smooth transition for you and your facility after the FMI installation.

If you're upgrading to Version 9.0 from 8.3, you will receive the following new features and options:

- Body Pak Software from ASP2: (FRFSE, Dual Echo GRE, Radial Tool, SSFSE-XL) PSD type-ins are no longer required. Select these sequences from the PSD menu.
- 3-Plane GRx
- Smartprep/Step Enhancements (Quiet Delay, Single click Prescan All Functionality, New Protocols)
- SSFSE-XL Enhancements (Flexible TR, TE and RBW)
- Variable Flip Angle T1
- T1 FLAIR
- MT compatibility with SE and GRE
- Flexible X RES and Zip512 selections for SE and 2D GRE
- VSS Sat Pulses with 3D CSI (Spectroscopy)
- Dynamic R1 for 3DFSE
- SCIC
- VSS Pulses for sites with Spectroscopy
- High B-Value Diffusion

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- SSFSE-XL Enhancements (Flexible TR, TE and RBW)
- Variable Flip Angle T1
- MT, 512 ZIP, Variable Frequency matrix compatibility with SE and 2D/3D GRE
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The Body Pak key must be re-installed by your service engineer after the 9.0 FMI is complete. All protocols utilizing Body Pak PSDs will need to be rebuilt as well.

The following 9.0 options are available for purchase:

- PROSE
- Fluoro Triggering
- 2D/3D Fiesta
- I-Drive Pro Plus
- Spiral Real-time and High Resolution Cardiac Imaging
- Delayed Enhancement
- Real Time Functional MRI

3-Plane Graphic Overview

The screenshot displays the 3-Plane Graphic Overview interface, which includes three main image windows and a control panel on the left.

- Top Left Window (Sagittal):** Shows a sagittal MRI slice with a red line indicating the phase FOV. A red dot on the line is labeled "Add Handle".
- Top Right Window (Coronal):** Shows a coronal MRI slice with a red line indicating the phase FOV. A red dot on the line is labeled "Rotate Handle".
- Bottom Window (Axial):** Shows an axial MRI slice with a red line indicating the phase FOV. A red dot on the line is labeled "Accelerator Line".
- Control Panel (Left):**
 - 3 Plane GRx:** Includes buttons for "Erase Group", "Erase All", and "Reset Center".
 - Tracker:** Includes checkboxes for "Report Cursor", "Keep W/L", and "Loc Ref Lines".
 - Auto Update:** Includes checkboxes for "W/L", "Pan", and "Zoom".
 - Phase FOV:** A slider set to 0.75.
 - Freq DIR:** A dropdown menu set to A/P.
 - FOV:** A dropdown menu set to 16.
 - Slice Thickness:** A dropdown menu set to 3.0.
 - Spacing:** A dropdown menu set to 1.5.
 - Accept:** A button at the bottom.

Annotations with arrows point to specific features:

- Reduced Phase FOV:** Points to the Phase FOV slider.
- Add Handle:** Points to a red dot on the Sagittal phase FOV line.
- Rotate Handle:** Points to a red dot on the Coronal phase FOV line.
- Accelerator Line:** Points to the red line in the Axial window.

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With this software release there has been a major change to the Graphic Rx screen, allowing you to see your graphic prescription in all 3 planes and do multiple obloquies in 1 prescription. The window now opens displaying 3 images at all times, a 3 Plane Localizer being the default.

What you need to know to get started:

- The last 3 Plane Localizer scanned is always loaded. Placing the center slice from each plane in each of the view ports.
- If there is no 3 Plane localizer the last compatible series scanned is loaded. The center slice is placed in the top right hand view port.

To change a image

- Click on the image you want to change, this places a yellow border around the image making it the active window.
 - o Then click on the “- or +” buttons to move through the images.
 - o Type the image number on the accelerator line.
 - o Click on select image and select an image from the list.

To deposit slices

- To deposit any group of slices (even the 1st) hold down the “shift key” and left click with the mouse.
- Slices can be added from either end of the group.
- The active slice group is blue, inactive groups are yellow.
- Slices can be deposited on any of the images.
- Once the slices are deposited they can be manipulated on any of the windows.

Phase FOV

- Slice lines lengths show any changes in the Phase FOV.

SAT bands

- SAT bands have to be deposited on the SAT screen, but show up on the Grx screen and can be manipulated there if needed.
- To deposit a SAT band, click on the “L,R,A,P,S or I” then hold down the “shift key” and left mouse click.
- Active SAT bands are red and inactive SAT bands are yellow.

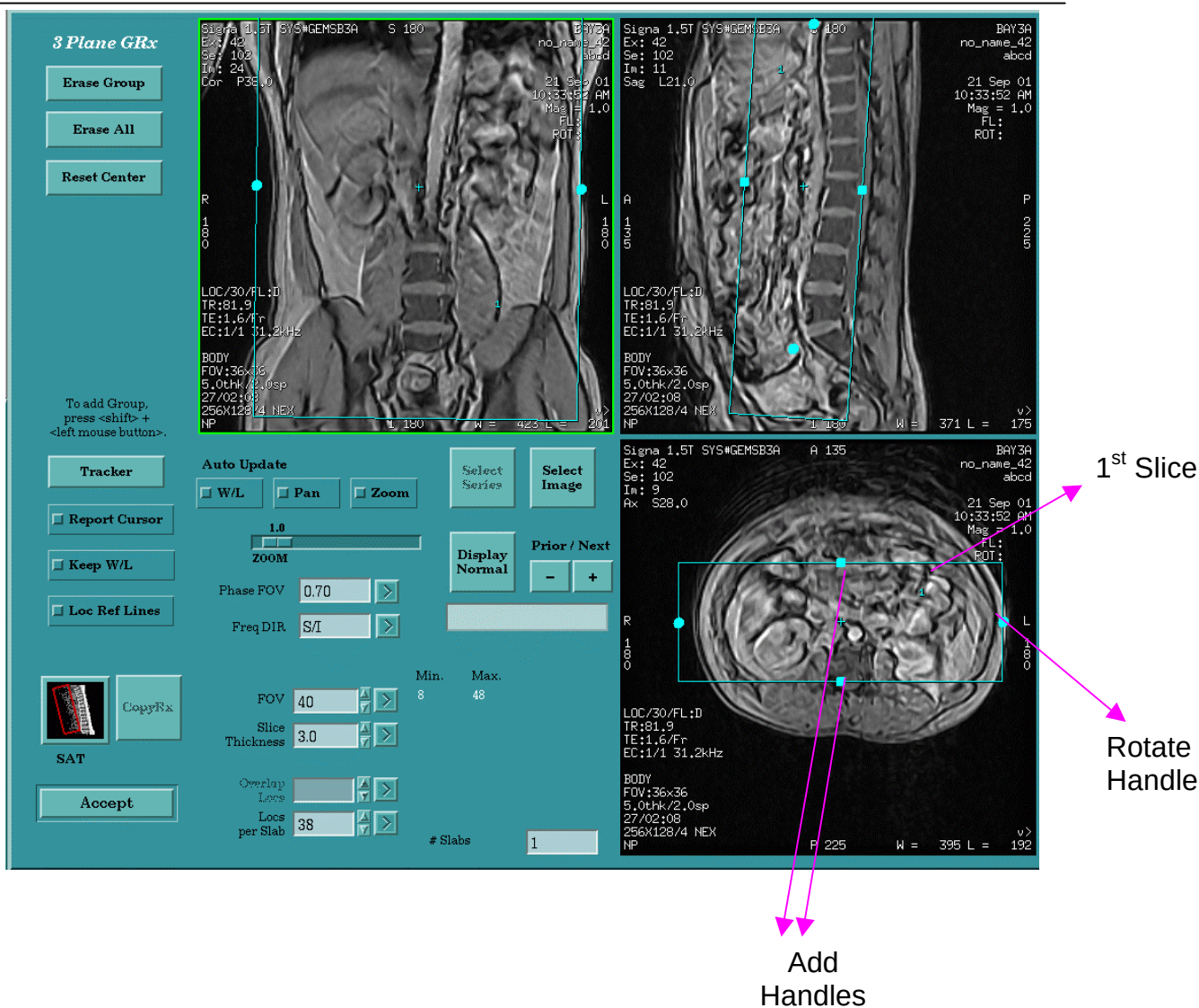
Reset Center

- This button is a new addition for most users, it enables you to deposit slices on one image then move through the slice and select another image as the center.
- For example, if scanning a shoulder, deposit the slices on the slice where you see the supraspinatus then move to the slice where you see the joint and click on “Reset Center”. This moves the Center of the FOV to that slice and maintains the angle.

Copy Paste

- Copy paste always calls up the images that were originally used to set up the prescription.

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3D Prescriptions

- Slabs can be added for either side.
- The number in the box is placed closest to the first slice in the volume.

For further details refer to the 9.0 New Features Manual.

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Smart Applications Tip!

Do not utilize a MIP projection from an axial TOF in combination with the fallback key in 3-plane GRx. This selection can result in table movement errors. Selecting fallback will cause the system to fallback to S/I zero. To correct for left/right and or A/P offsets, deposit your volume or slice locations on a source image when prescribing in 3-plane GRx and utilize both the MIP and source images to make adjustments in slice or volume location.

Copy Rx – always uses the original series to post the slice locations. It will not allow the slice location to be prescribed on another series. The ability to Copy Rx onto another series will be added to a future software release.

The following message may be seen occasionally “Communications with the graphic server lost, GRx prescription will be closed. Please reopen to continue”.

This message represents an orderly shutdown of the GRx and helps to prevent system crashes. When GRx is re-opened, your slice Rx should still be available, if not, re-deposit them and proceed with scanning.

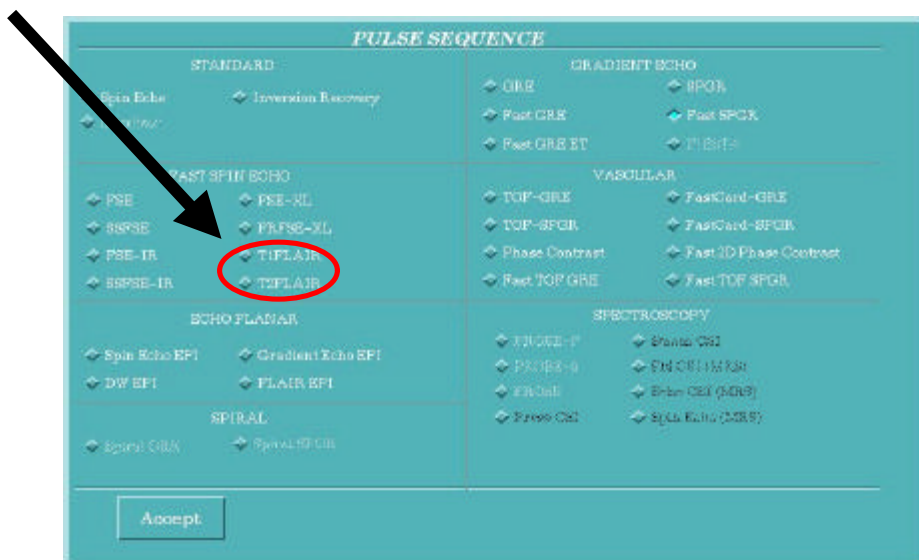
New PSD Selections:

T1 FLAIR: Parameter selections are now flexible for both “Picture This” and Site Protocols. To activate, select FLAIR T1 from the PSD menu.

T2 FLAIR: This option is now available on the PSD menu, and replaces the old “FLAIR” selection

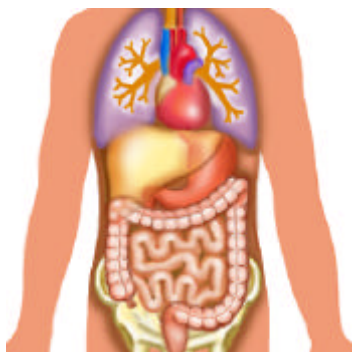
SSFSE-XL: There are multiple changes to this PSD. Please review the SSFSE-XL section in this document and the Version 9.0 Operator Documentation for more information.

FRFSE-XL: Parameter selections are now flexible for both “Picture This” and Site Protocols. To activate, select FRFSE-XL from the PSD menu.



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Body Pak Pulse Sequences



- **FRFSE-XL**- Offers improved resolution and T2 contrast. A PSD type in is no longer required to activate this pulse sequence. Select FRFSE-XL from the PSD menu to activate this sequence.
- **Dual Echo FGRE**- Acquires both in and out of phase images in a single breath-hold. There is no longer a PSD type-in for this sequence. Select either FGRE or FSPGR from the PSD menu and 2 echoes on the timing screen to activate.
- **High Resolution MRCP with SSFSE-XL**- Offers flexible acquisition matrix selections, radial tool option and multi-angle slice prescription. **Select SSFSE-XL from the PSD menu to activate.**

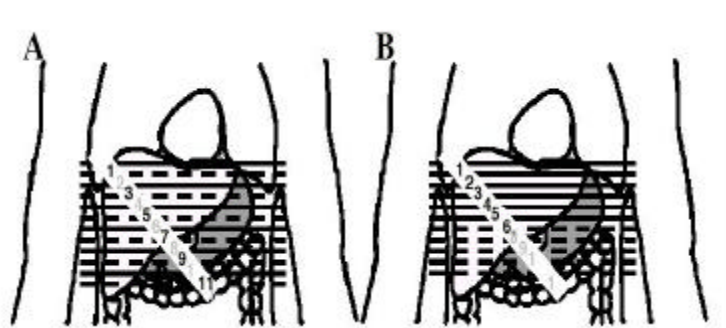


Figure A represents the “old” slice ordering scheme with FSE T2 weighted imaging. Odd slices were acquired during the first breath-hold. Even slices were acquired during the second breath-hold.

Figure B represents the improved slice ordering scheme applied with FRFSE-XL. Images one through 6 are acquired in an interleaved fashion during the first breath-hold, and images 6-11 are acquired in an interleaved fashion during the second breath-hold. The middle slice (slice 6 and slice 7) are prescribed at the same location to provide an overlap.

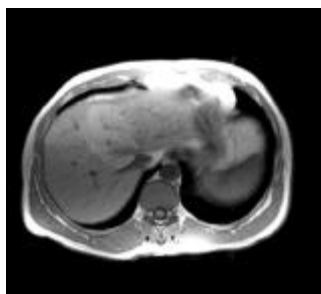
Blurring Cancellation is a new user CV compatible with FRFSE-XL. This new option acquires k-space twice in reverse order to reducing ghosting artifacts. Blurring Cancellation is not compatible with .5 NEX.

For more detailed information regarding FRFSE-XL please refer to the ASP2 operator documentation.

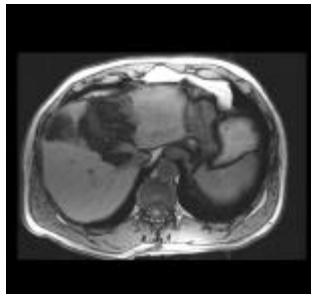
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Dual Echo FGRE

This pulse sequence will acquire both in and out of phase images in a single breath-hold. Total acquisition time is reduced by $\frac{1}{2}$, and scan location discrepancies between each breath-hold are eliminated.



In Phase



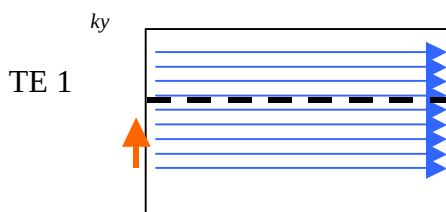
Out of Phase

Single Shot Fast Spin Echo

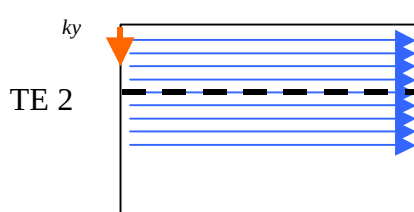
Version 9.0 SSFSE now offers increased flexibility and improved SNR. You will notice that SSFSE protocols now allow user selection of any value to be entered in the TE and TR fields. On previous software releases parameter selections in these fields were restricted.

TE choices are no longer limited to the TE values listed on the pull down menu. Below you will find recommendations for selecting TE values for each field. For more information regarding “Reverse Linear” and “Linear” View Ordering, please refer to the Version 9.0 New Features Manual.

- Select a value listed for TE1 when performing **standard SSFSE** imaging. Any TE selection between the min and max values listed in the advisory panel may be entered in the TE1 field. This will ensure that “**Linear View Ordering**” is utilized which is the recommended phase encoding scheme for short to medium range TE values.
- Select a value listed for TE2 when performing **MRCP** imaging. Select a TE value 10-15 ms longer than the minimum value listed in the advisory panel. This will ensure that “**Reverse View Ordering**” is utilized which is the recommended phase encoding scheme for long range TE values.
- With certain matrix combinations you will notice gap between the TE1 and TE2 ranges. Selection of a TE value within this range will result in download failures.

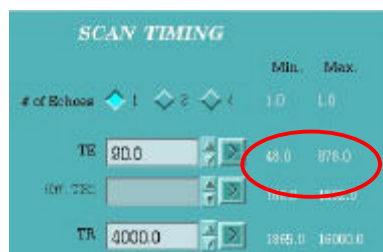


Linear View Ordering



Reverse View Ordering

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TE1 Range
Linear View Ordering



TE2 Range
Reverse View Ordering

Issue 1: A TE of 180ms will produce a very noisy image.

Solution: Use a TE of 160ms, you will obtain better image quality.

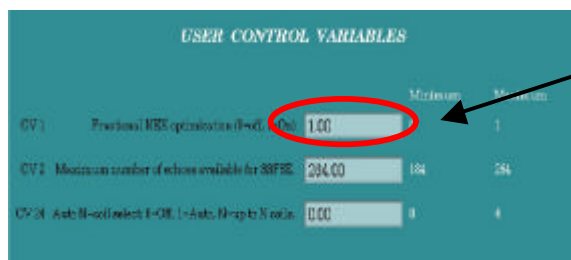
Smart Applications Tip:

When performing **MRCP** exams, you may need to adjust your YRES to obtain the desired TE values for reverse phase encoding (**10-15 ms longer than the minimum value in the TE2 advisory panel**).

TR choices are also flexible and are annotated on the image differently. In prior software releases, the TR annotated on the image represented the total time required to complete all slices in a series. In the Version 9 software, the TR annotated on each image represents the time required to complete only one slice of a series.

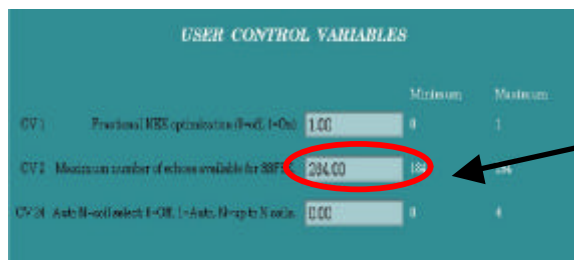
- To acquire all slices in the series with approximately the same TR value as before, select **Min TR**.
- To pause between each slice, **input a TR value other than minimum**. If a TR value other than Min is entered, the system will acquire the first slice and then pause during the remaining portion of the TR before acquiring the second slice. You may wish to utilize a longer TR to create a pause between slices. Creating a pause between slices results in increased tissue relaxation between each slice. For example, if you enter a 4000ms TR, the slice will take just under a 1 second to scan, and the system will pause for 3 seconds. This can be especially helpful in increasing the contrast in brain imaging.

SSFSE-XL User CVs



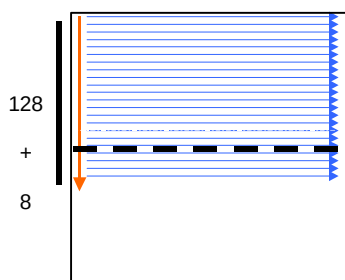
Fractional NEX Optimization
"on" will acquire the additional echoes that would normally be discarded. This will increase the number of over-scans, and increase the SNR. Only compatible with long TE scans.

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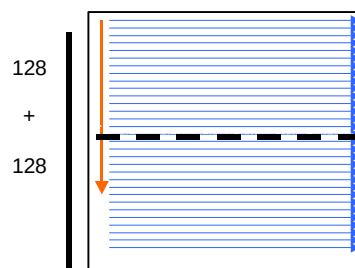
Maximum number of Echoes allows the user to choose the maximum number of echoes that will be available for over-scans.

Reverse View Ordering



.5 NEX

Reverse View Ordering



1 NEX

- Fractional NEX Optimization **OFF**- 256 Phase Matrix

- Fractional NEX Optimization **ON**- 256 Phase Matrix
- Maximum number of echoes: 256

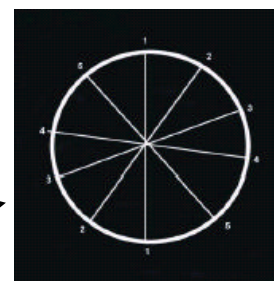
SSFSE-X Radial Tool



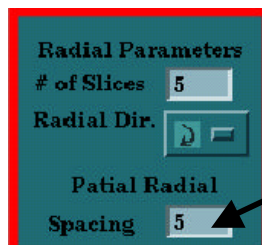
The Radial tool is a new option for SSFSE-XL that will allow graphic prescription of full radial or partial radial scans in a clockwise or counter-clockwise direction.



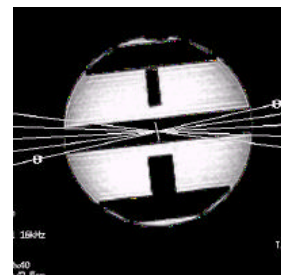
To prescribe a “wagon wheel” scan, simply enter the number of slices required, and select “0” for the partial radial spacing. Your scan prescription should look like this.



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To create a “cat whiskers” scan prescription, type a value in the partial radial spacing field. Your scan prescription will look like this.



To add another radial Rx, or to add another group of slices, **hold the shift key** down and place the cursor on the images where you would like to begin prescription. You may repeat as necessary. For more information regarding the radial tool, please refer to the ASP user documentation.

Fast TOF GRE/SPGR (Contrast Enhanced Angiography)

Version 9.0 also offers an enhancement to both the Smart Prep, and the Fluoro Triggering imaging options known as **Quiet Scan Delay**. Now the entire scan delay time, except for the last second, is silent. When performing CEMRA, the scan delay time is typically used to deliver patient breathing instructions. This enhancement allows easier communication between the patient and the scan operator. Scan delay times may also be entered even if you are not using Smart Prep.

In addition, the k-space filling order is now flexible with any Fast TOF sequence independent of the method used to initiate the sequence. For example, “centric” k-space filling (or any other k-space filling scheme) may be selected without Smart Prep.

SmartPrep and SmartStep

New for SmartPrep and SmartStep with Version 9.0!

- Scan Rx can be saved with Meta series
- Quiet Acquisition Delay
- Automatic Prescan for all stations
- Mask and Venous scans are now listed as a user CV
- Left Mouse clicks **only** are used to activate all scan functions including “Auto Step”
- Meta series can be copied and pasted



Prescan All button will perform prescan for each meta series

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Turbo Mode

Turbo mode is now a user CV for Fast TOF and Fast GRE/SPGR. This option, available on High Speed, Echo Speed and Twin Speed systems, reduces the width of the RF pulse to allow shorter TR periods and thus shorter total scan time.

- Turbo Mode **off** - RF duration **1.6ms**
- Turbo Mode **on Faster**- RF duration **.8ms**
- Turbo Mode **on Fastest** – RF duration **.6ms**

Smart Applications TiP!

Any image may be used for the volume or tracker placement when graphically prescribing a Smart Prep exam. The tracker will only be seen on the original image on which it was prescribed, and thus can only be manipulated on that image. Adjustments to the volume can then be made on any of the other 2 planes.

Issue: The number of stations can't be changed in a saved protocol.

Solution: If a different number of stations are to be scanned, a completely new scan protocol must be created and saved to site protocols. GE protocols cannot be changed.

Fluoro Trigger

CV19 – Real time SAT (0=Non axial, 1=axial) - This CV determines where the SAT bands are located during the real time acquisition. Before beginning your real time acquisition, you will need to determine the plane to be used to monitor the contrast arrival.

- If a sagittal or coronal plane is selected to view the contrast arrival, enter the default value of **"0"** or non axial. This selection places the SAT bands at the edge of the scan FOV during the real time scan.
- If an axial plane is selected to view the contrast arrival, enter a value of **"1"**. This selection places the SAT bands parallel to the slice during the real time scan, and allows them to move with the slice.

Smart Applications Tip!

Many sites have found it most effective to an axial real time scan for the carotid arteries, and a sagittal or coronal real time scan for the renal arteries.

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K-Space Filling

When utilizing Smartprep or SmartStep imaging options, there are several ways to fill k-Space. The following four options can be found on the user CV page. In addition, these methods of K-Space filling are no longer linked to Smartprep / SmartStep. Centric and Elliptic-Centric can now be utilized with any FGRE or FSPGR scan prescription.

K-space Filling Options	Fill K-space How...
Reverse Elliptical-centric	Fills outer lines and outer slices first.
Elliptical-centric	Fills the center lines and center slices first.
Centric	Fills the center lines first.

K-space Filling Options	Fill K-space How...
Reverse Centric	Fills the outer lines first.

Please refer to the Version 9 operator documentation for explicit details regarding scan set-up for SmartPrep and SmartStep.

Gradient Echo

Mag Transfer is now compatible with 2D GRE. There is also a new user CV called "Mag Transfer Pulse Type" which represents the duration of the Magnetization Transfer Pulse.

- Select MT Pulse **1** for 8ms duration
- Select MT Pulse **2** for 16ms duration
- For "MT offset" use the default value.

Spectroscopy

Version 9.0 Spectroscopy users will notice several changes to scan RX and spectrum display.

The post processing of spectroscopy data with AW Functool, MIROI, or Version 9.0 FuncTool, still requires that "**probeSI**" be typed into the psd name field.

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Prescriptions Changes:

- Phase, Frequency, and number of slices will need to be filled in for all spectroscopy scans. The chart below will help guide you regarding proper selection of these values.

Spectroscopy Imaging Mode	Phase	Frequency	NEX	Number of Slices
Single Voxel	1	1	Represents the phasing has no affect on scan time	1
Multi Voxel (2D CSI)	8 to 32	8 to 32	Usually 1, anything higher will increase scan time	1
3D Multi Voxel (3D CSI)	8 to 32	8 to 32	Usually 1, anything higher will increase scan time	8 to 16

Display Changes:

- In prior software versions (8.3), all 2D CSI series were zero filled to display the spectral information as a 32 square matrix. With Version 9.0 software, the images will now reconstruct to the closest factor of 2.

Phase or Frequency Selection	Reconstruction Display
8	8
10 to 16	16
16 to 32	32

Graphic changes for Spectroscopy:

Spectroscopy sequences are the only place that single plane graphic is still used. There are now two new buttons within graphic, "Start" and "End". These keys can be used in two different ways.

- Deposit the VOI on the image by left mouse clicking anywhere on the image, then click on the "Start" and "End" keys with the same image on the screen. This will set the image

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- on the screen as the center of the slice or volume maintaining the slice thickness typed in the slice thickness field. The same as previous software.
- Deposit the VOI on the image that is the beginning of the region of interest, click “Start”, then move to the image at the end of the region of interest and click “End”. This will set the boundaries of the volume and overwrite the slice thickness field to reflect the start and end locations.

When paging through slices with the VOI deposited, the “Within and Out of Boundaries” information can be confusing. It is only accurate if “Start” and “End” have both been clicked.

Issue: Once you have paged “Out of Boundary” the “Within Boundary” message doesn’t turn back on, on the correct slice.

Solution: Rely on the Start and End locations in the scanning range information.

Issue: The Version 9.0 video states that oblique imaging is not compatible with 2D/3D Spectroscopy.

Solution: This issue was resolved after production of the Version 9.0 video. Oblique imaging is now compatible with Version 9.0 software.

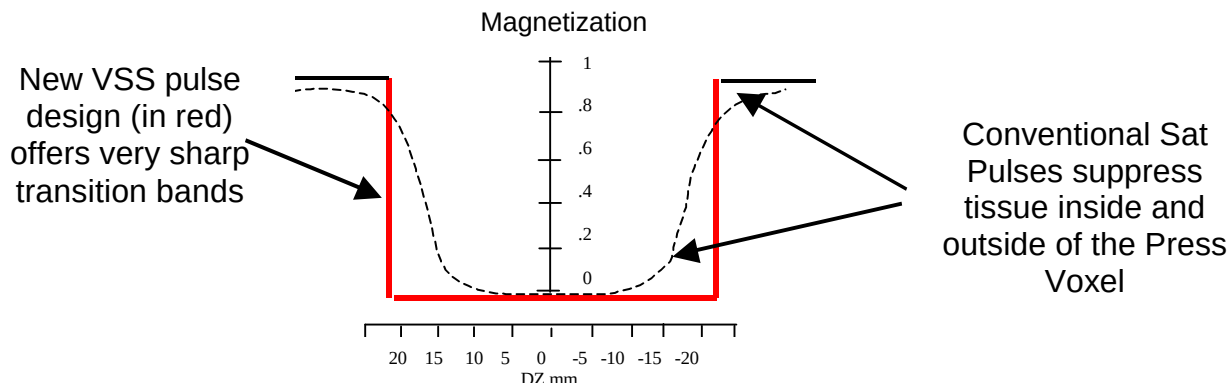
Issue: In Spectroscopy protocols saved in 8.3, the “tempC” CV may get set to the wrong temperature, resulting in the Spectra been shifted on the ppm scale.

Solution: Rebuild the protocol. This will reset the CV.

VSS Pulses

“Very Selective Sat” pulses are new for Single Voxel, 2D or 3D Probe-P applications with Version 9.0 software. This new design offers the following advantages:

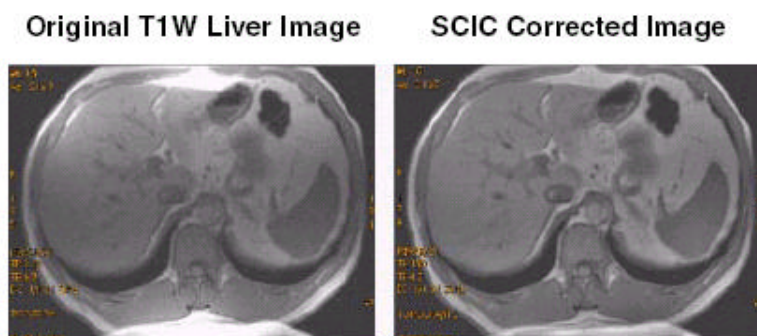
- Six very precise sat pulses can be applied automatically at the edge of the probe voxel.
- Up to four additional sat pulses can be applied manually in Grx.
- Provides excellent suppression of unwanted signal outside of the spectral volume
- Offers tremendous improvement in overall quality of the spectral display



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SCIC- Surface Coil Intensity Correction

This option provides correction for signal inhomogeneity on phased arrays coils. You can select SCIC from the imaging options menu.



SCIC is compatible with all imaging planes and pulse sequences, but is only available with the coils listed in the chart below.

	1.0 and 1.5T Coils	1.0T Coils Only	1.5T Coils Only
USA Instruments CTL Spine Coil	USCTLTOP, USCS123, USCS12		
	USCTLMID, USCT284, USTL345, UST623, UST634		
	USCTLBOT, USTL45, USLS456, USLS56		
GE by Teledyne CTL Spine Coil	CTLTOP, CS123, CS12,		CT234
	CTLMID, TL345, TS34, TL45	TL456	TS23
	CTLBOT	LS567, LS56, LS67	LS456, LS56
Other Coils	ENDOREC		
	TORSOPA		
	PVUPPER		
	TPUPPER, TPLOWER		

Variable Flip Angle T1

Version 9 software now offers a variable flip angle to optimize T1 contrast. To access this new feature, you will need to type “**t1memp**” in the PSD line. The chart below provides suggested flip angle selections based on the selected TR value

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TR Value Optimal Flip Angle

200	100
300	88
400	80
500	73
600	68
700	63
800	59
900	56

High B-Value Diffusion

High B-Value DW- EPI provides the following improvements to diffusion weighted echo planar imaging:

- B-Values up to 2,500 for Smartspeed
4,000 for High Speed Plus
7,000 for Echo Speed Plus
- Minimum TE based on the B-Value selected
- Multi- NEX averaging



Optimize TE is defaulted "ON". This option will allow increased B-Value selection. If Optimized TE is not selected, B-Values will be limited to 1000.

Optimize TE will calculate and utilize the shortest gradient duration possible to achieve the selected B-Value, this in turn will result in shorter TE values.

For more information on High B-Value Diffusion, please refer to the ASP2 operator documentation.

Dynamic R1

Dynamic R1 is an enhancement to the 3DFSE pulse sequence designed to improve SNR for large FOV scans by increasing the receiver values at the edge of K-Space. This reduces digitization noise and increases overall SNR.

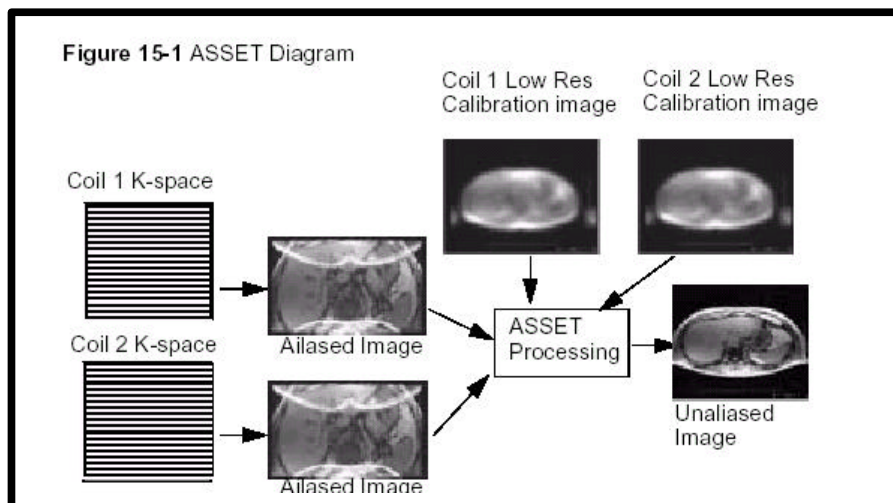
To access Dynamic R1, select a 3DFSE pulse sequence and type "**3dfsedr1**" on the PSD line,

Asset

Asset is a new imaging option available with Fast GRE and Fast TOF pulse sequences with offers reduced scan time with phased array coils. This option works in a similar manner to 0.5 phase FOV in which every other line of k-Space is filled to provide the same resolution with a

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50% reduction in scan time. To prevent wrap artifacts, Asset requires that a calibration scan be performed on each component of the phased array coil to allow the system to account for anatomy outside of the PFOV. For more information on Asset, please refer to the Version 0 New Features Manual.



Asset is currently only compatible with the Torso Phased Array Coil

Error Messages

Issue: “Advisory Error” (Pink) pop-up message, and a “Coil, Patient Position” (Blue) message will appear on operator screen at the same time. These messages may also be displayed one on top of the other resulting in what seems to be a system lock-up.

Solution: The “Advisory Error” message has to be OK’ed first. To access the “Advisory Error” message, move the “Coil, Patient Position” message out of the way.

Issue: “Archive media detached” message in the system status window.

Solution: This message represents a normal system operation. With Version 9 software a fail safe was added to detach the MOD when the system remains idle for more than one hour. This enhancement prevents corruption of the MOD in case of power failures or system errors. When auto archive has been activated the MOD will resume archiving when a new exam has been completed. If auto archive has not been activated the system operator will need to initialize the archive function after the completion of the exam.

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Your LX system recently underwent a Version 9.0 software installation. In order to address several issues that have been brought to our attention, your Field Service Engineer has loaded a series of software updates. These software updates include fixes to the following:

- Changes in display to prevent
 - o Images displaying in the Viewer out of order if the series was displayed before reconstruction was complete.
 - o The reference image not displaying during multiple acquisition scans.
 - o Image jumping when performing a cross reference in the viewer.
- 3 - Plane Graphic intermittently displayed images from the previous patient.
- With GRE sequences saved with a software other than 9.0, a message was displayed "ihoffsetfreq is out of range". Your FE will also run a script to update all the Site and GE protocols.
- Changes to the Coil Config manager to allow 2 new coil configurations to be added to your system ENDOATD and ATD3ENDO (1.5T only). These coils are used primarily for prostate spectroscopy.
- Reformat: The reformat menu closes while in "Batch" mode creating thick slice MIP images.
- Archive and network problems after loading of service software.
- Returning the Elliptic centric ordering selection for all 3dfgre/spgr sequences.
- The removal of 2 nuisance messages "opmonthick is out of range" and "Unable to open HostVMSampleData.dat file"
- FuncTool will now process 2D CSI images. The software enhancements and reinstallation of the FuncTool will remove the FlexTrial version of the FuncTool CSI. The FuncTool CSI key is only needed to post process 3D CSI data .
- Changes in image contrast on Cine Phase Contrast images, due to incorrect mask CV defaulted value.
- For 1.5T TwinSpeed systems only. After an amplifier trip the scanner was scanning with the wrong gradient coil, resulting in images that appeared as if they were scanned at the wrong FOV.
- Smartprep tracking failures related to restrictions in tracker and volume placement and Smartprep failures related to early tracker detection issues.
- The respiratory triggering option fails to recognize the prescribed trigger window, resulting in data acquisition throughout the entire 2-RR period in FSE, SSFSE and FRFSE.
- Copy paste feature within the RX manager is now available with 2D Fiesta sequences. In addition, FOV selections are now flexible and double oblique prescription is compatible. 2D Fiesta is now available as a separate option from 3D Fiesta.
- SSFSE image quality issues related to poor or inconsistent fat sat or failure of fat sat during manual prescan.
- Echo Planar image ghosting issues on Twin Systems, and the appearance of "dot" artifacts on DW-EPI.
- Oblique 2D / 3D Spectroscopic imaging is now compatible with Version 9.0 software.
- Image quality issues related to the use of fractional NEX optimization with SSFSE which presented as image wrapping and elongation.
- Image rotation or flipping issues related to SSFSE MRCP exams performed with MSMA. (This issue has been resolved since the production of the Version 9.0 video. You will no longer be required to flip or rotate the images in the viewer to correct the image display.)

g LX Version 9.0 Applications Update

- MRA exams post processed with IVI on a calendar day other than that of data acquisition will display the scan acquisition date and not the post processing date.
- Patient data processed within the HIS/RIS application will no longer combine exam or image data from multiple patients.
- 3DFSE PSD spontaneously aborts the second slab in the scan acquisition.
- Lipid peaks are no longer shifted and are displayed accurately when viewing the spectral data in “spectro pre-scan”.
- PROSE protocols are now updated in the GE Protocol Library.
- PROSE auto-prescan values are now accurate and calculated only from the endo-rectal coil. When performing PROSE imaging, the “combination coil” should only be used for imaging larger FOVs to properly evaluate outlying anatomical structures with standard MR imaging as part of the total prostate work-up. Use of the endo-rectal coil should be reserved for dedicated spectroscopic imaging of the prostate.
- EPI images will now display the pulse sequence annotation.
- The gradient shut down issue (no gradient noise) during the baseline phase of Smartprep data acquisitions has been corrected.